Experiment: Clustering using K-Means and Hierarchical Clustering

Aim

To implement and compare **K-Means Clustering** and **Hierarchical Clustering** on a sample dataset using Python.

Theory

1. K-Means Clustering

K-Means is an unsupervised machine learning algorithm used to group data into **K clusters** based on similarity.

Steps:

1. Choose number of clusters (K)
2. Initialize K centroids randomly
3. Assign each data point to nearest centroid
4. Recalculate centroids
5. Repeat until convergence

Key Idea: Minimize within-cluster variance.

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2. Hierarchical Clustering

Hierarchical clustering builds a **tree-like structure (dendrogram)**.

Two types:

- **Agglomerative (bottom-up)** ← commonly used
- **Divisive (top-down)**

Agglomerative Steps:

1. Treat each point as a cluster

2. Merge closest clusters
3. Repeat until one cluster remains

Linkage methods:

- Single linkage
- Complete linkage
- Average linkage
- Ward's method

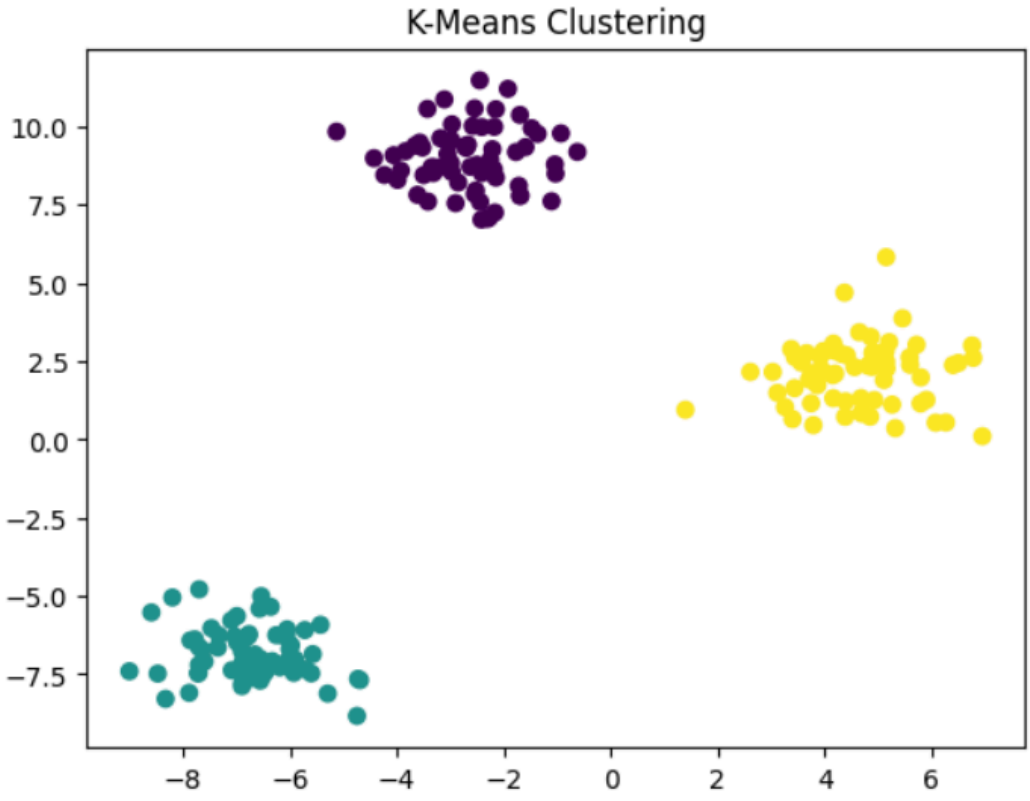
Procedure

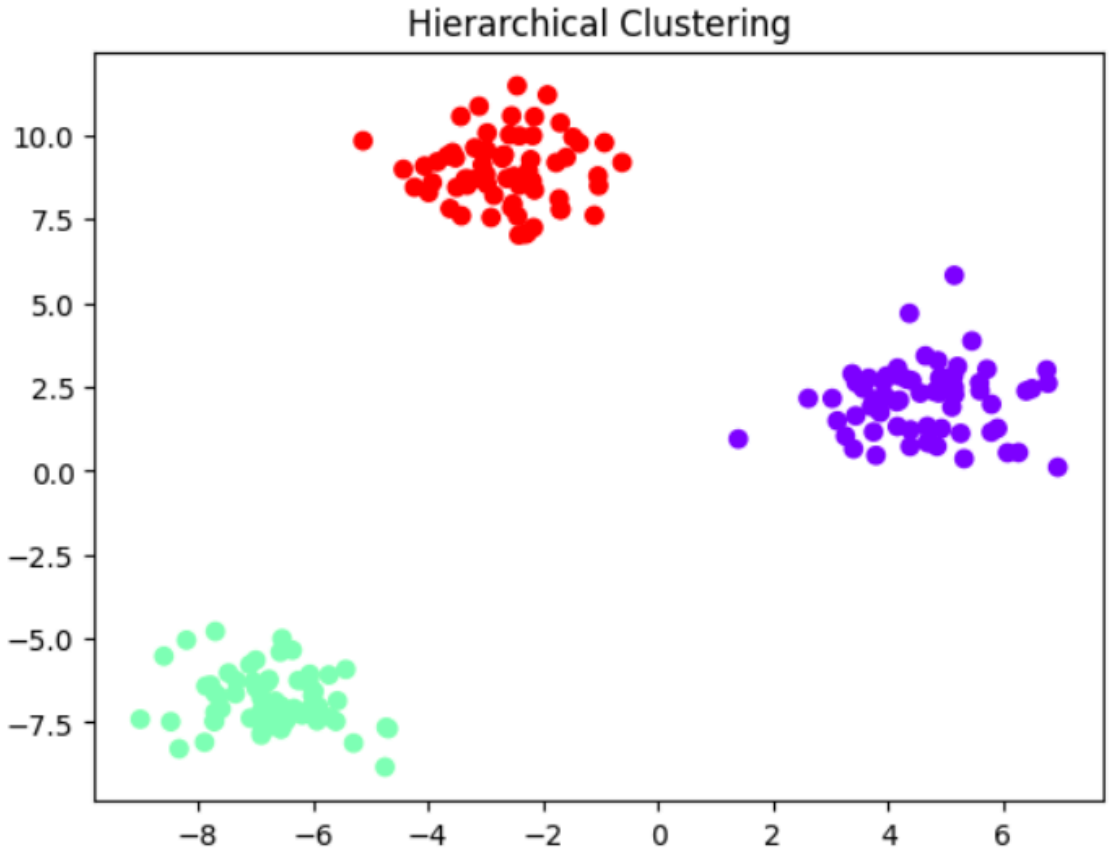
1. Import required libraries
2. Load or generate dataset
3. Preprocess data (if required)
4. Apply K-Means clustering
5. Apply Hierarchical clustering
6. Visualize results
7. Compare outputs

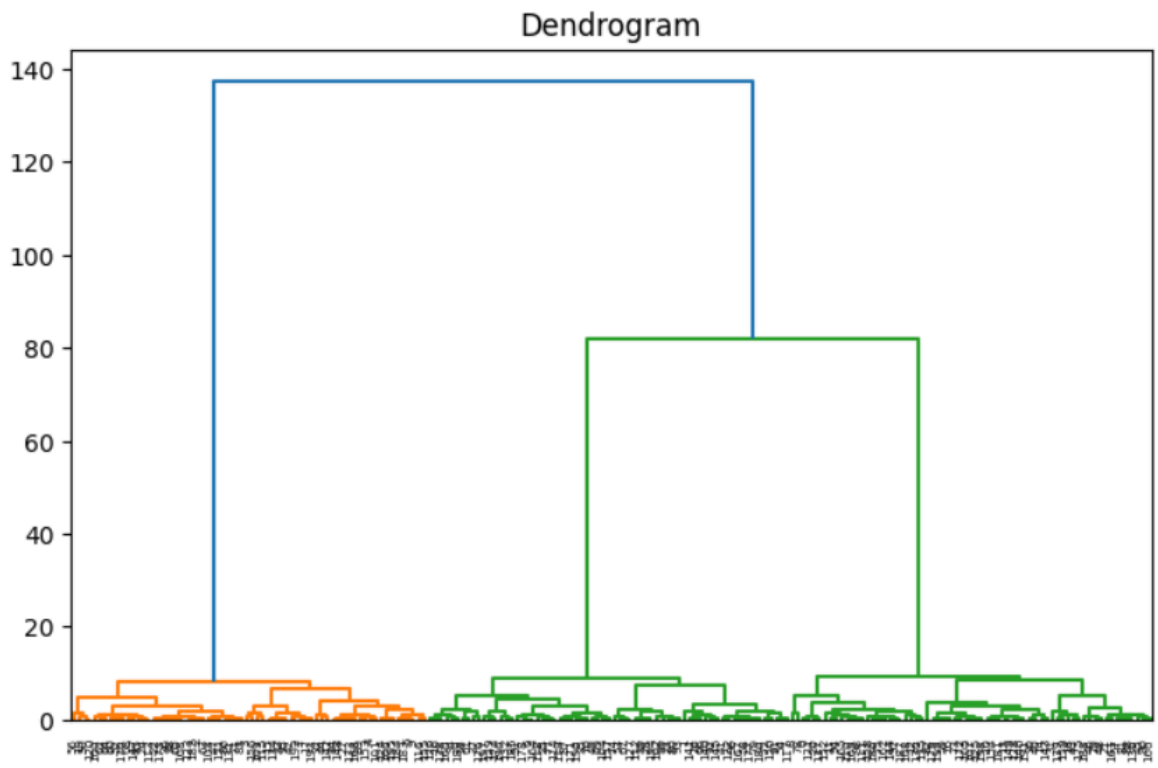
```
# Import libraries
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage

# Step 1: Create sample dataset
X, y = make_blobs(n_samples=200, centers=3, random_state=42)

# Step 2: K-Means Clustering
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans_labels = kmeans.fit_predict(X)

# Plot K-Means result
plt.scatter(X[:, 0], X[:, 1], c=kmeans_labels, cmap='viridis')
plt.title("K-Means Clustering")
plt.show()
```

```
# Step 3: Hierarchical Clustering (Agglomerative)
hc = AgglomerativeClustering(n_clusters=3, linkage='ward')
hc_labels = hc.fit_predict(X)

# Plot Hierarchical result
plt.scatter(X[:, 0], X[:, 1], c=hc_labels, cmap='rainbow')
plt.title("Hierarchical Clustering")
plt.show()

# Step 4: Dendrogram
Z = linkage(X, method='ward')
plt.figure(figsize=(8, 5))
dendrogram(Z)
plt.title("Dendrogram")
plt.show()
```

OUTPUT




Hierarchical Clustering
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−8
−6
−4
−2
0
2
4
6



Conclusion

- **K-Means** is efficient and fast but requires predefining the number of clusters.
- **Hierarchical Clustering** does not require predefining clusters and provides better visualization through dendrograms.
- K-Means works well for large datasets, whereas hierarchical clustering is more suitable for smaller datasets.
- Both methods produced similar clusters on the sample dataset.